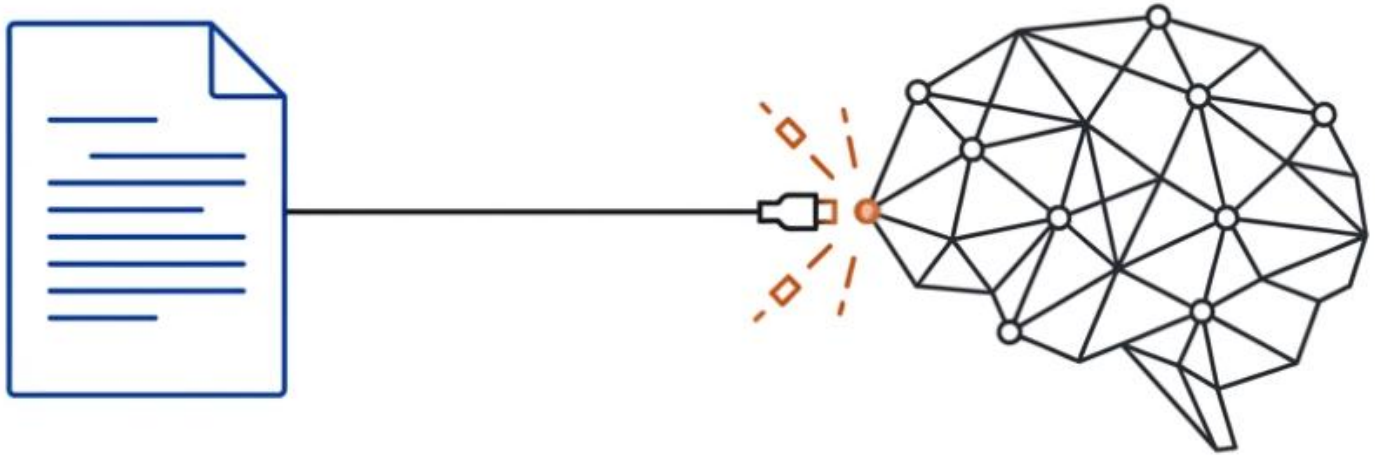
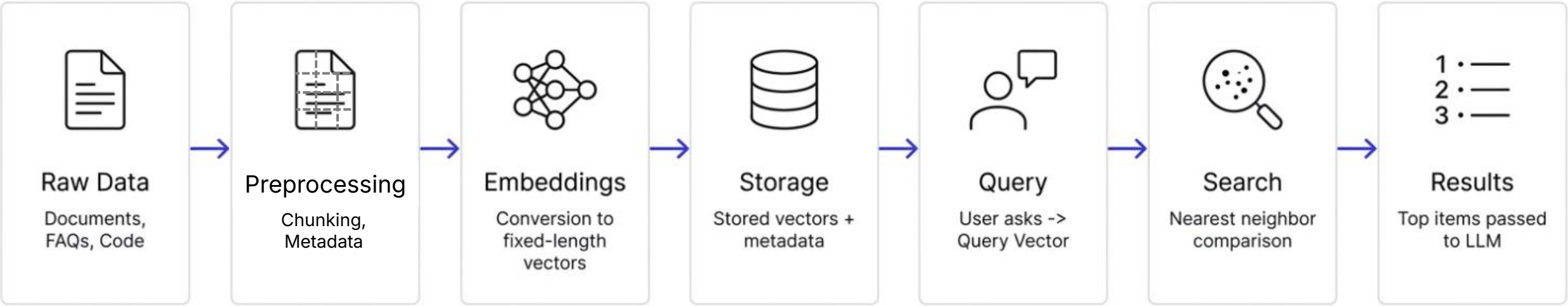


From Search to Answers: Building Your First RAG Pipeline

Ingestion → Chunk → Index → Retrieve → Answer.
LangChain/LlamaIndex

https://tinyurl.com/rag-augmented

The picture till now...



Even a perfect retriever leaves the heavy cognitive lifting to the user.

Fragmented Context

The complete answer is spread across multiple disparate chunks. Synthesis is missing.

Unfriendly Formats

Chunks are raw excerpts filled with partial paragraphs, technical phrasing, and incomplete sentences.

Evidence Dumps

Users face cognitive overload, forced to read, compare, and summarize the returned text themselves.

Missing Evidence

Retrievers return the 'best available' match, even if those chunks do not actually answer the prompt.

"Retrieval finds the evidence. The LLM turns evidence into answers.

Top-k chunks are not the final product; they are the input to the final product.

The LLM bridges the gap through four distinct capabilities.

1. Synthesis

Combines distinct information fragments from multiple retrieved chunks into a single, cohesive narrative.

2.
2. Natural Language Formatting

Converts raw, technical excerpts into pedagogically useful summaries, bullet points, or beginner-friendly text.

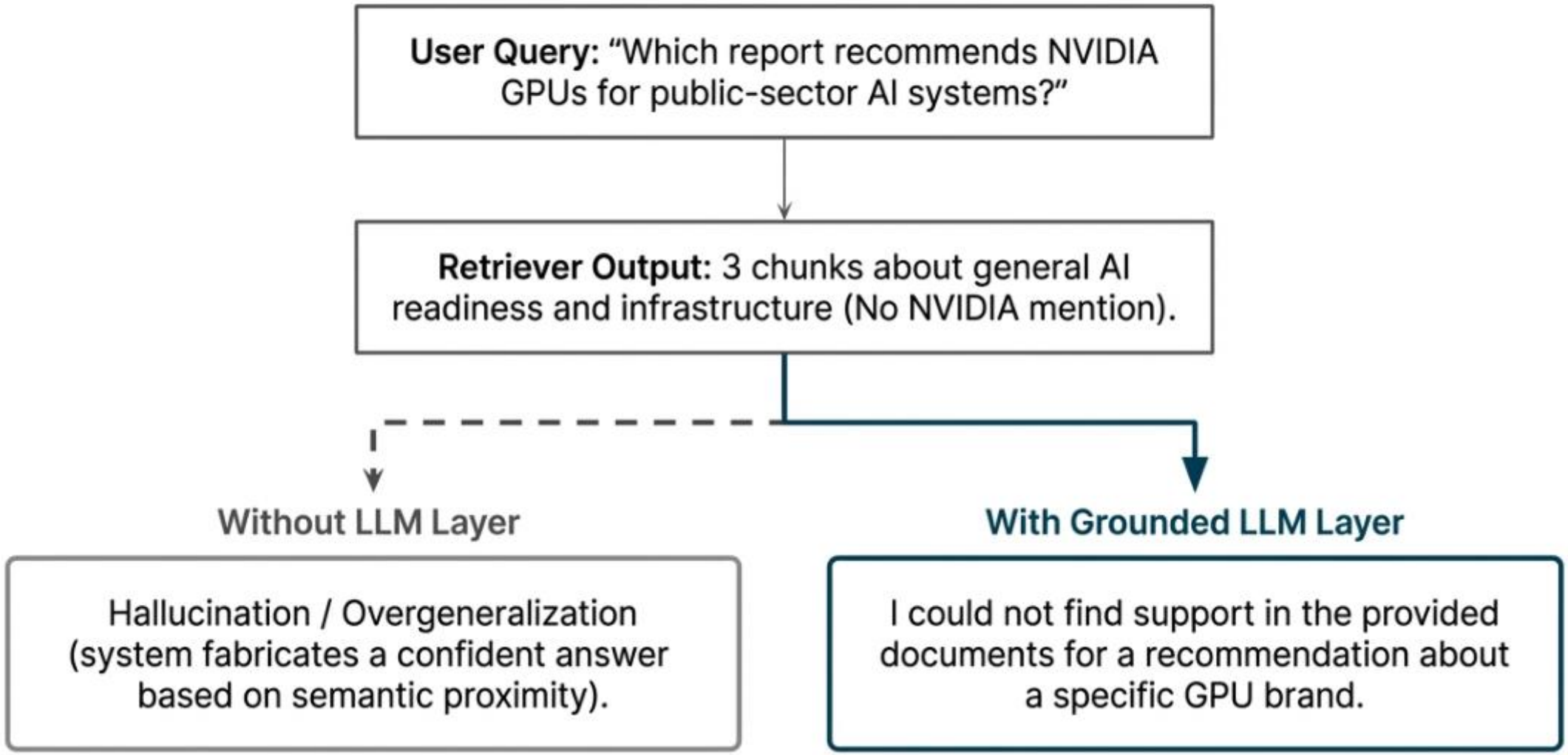
3. Citation Awareness

Grounds the final answer with source metadata (e.g., [OECD, p. X]) to build user trust.

4.
4. Answer Discipline (Refusal)

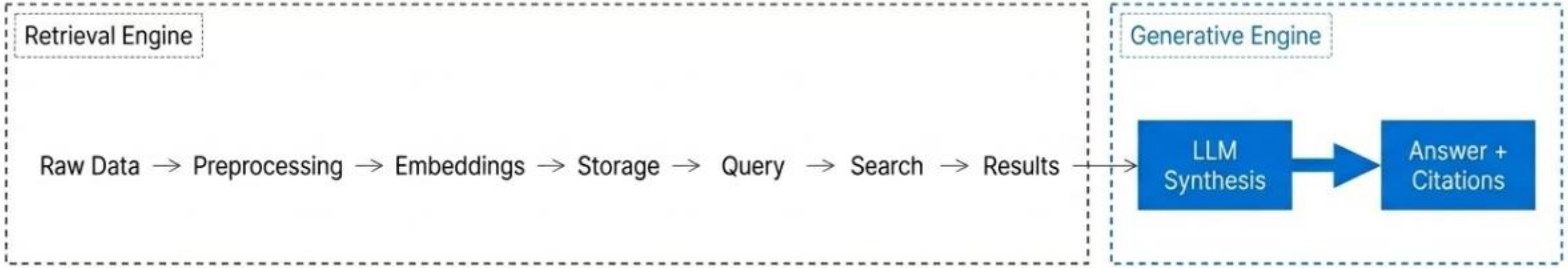
Exercises the discipline to say "I don't know" when corpus evidence is insufficient.

A grounded LLM turns missing evidence into trustworthy refusal.



The Full RAG Architecture Blueprint

A complete, end-to-end system for querying curated knowledge bases.



Retrieved Evidence

Chunk 1: AI governance requires public-sector capacity.

Chunk 2: Openness matters for transparency.

Chunk 3: Digital infrastructure supports scalable adoption.

Synthesized Answer

Across the reports, responsible AI adoption is presented as depending on three connected elements: governance capacity, openness in system design, and strong digital infrastructure foundations.

Lab

Thank You

Appendix